



Diffusion-based knowledge distillation for effective multi-organ segmentation with reduced computational time

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ABSTRACT

Accurate and efficient multi-organ segmentation is crucial for clinical workflows, requiring high accuracy and reduced computational time. In this research, we propose a 3D diffusion-based knowledge distillation framework (3DKD-DiffuseNet) for multi-organ segmentation to achieve higher accuracy with reduced computational time. The core idea is to enhance the training of a lightweight student model by integrating a diffusion mechanism that guides feature learning during knowledge transfer from a high-capacity teacher model. Unlike conventional distillation approaches that rely solely on soft label supervision, our framework incorporates a diffusion consistency loss that encourages the student to learn stable and spatially coherent representations. To further improve computational efficiency, we introduce an organ-specific intensity thresholding strategy, which localizes regions of interest and reduces unnecessary processing without sacrificing critical anatomical detail. The model is validated on both MRI and CT modalities for brain tumor segmentation (BraTS benchmark) and abdominal organ segmentation (RAOS) tasks. On the BraTS benchmark datasets, it achieved outstanding Dice scores for high-grade and low-grade tumors, outperforming the teacher model by 3%–5% across all modalities. On the RAOS dataset, it similarly delivered excellent Dice scores, with improvements of 3%–6% for critical organs compared to other state-of-the-art (SOTA) models. Experiment shows that the model also achieved a 2–3× reduction in computational time due to strategic preprocessing. Our diffusion-based student model, supported by strategic preprocessing, offers enhanced segmentation accuracy and computational efficiency, making it suitable for clinical applications that require fast and reliable analysis.

1. Introduction

Automated organ segmentation in three-dimensional (3D) medical imaging has the potential to enhance diagnostic accuracy, treatment planning, and disease monitoring [1]. Segmentation of vital organs and precise identification and delineation of structures, such as brain tumors, could aid clinicians in making accurate diagnoses and individualized treatment decisions [2,3]. Manual segmentation, however, is labor-intensive [4], exhibits significant variability among different observers [5], and may result in inconsistencies across extensive 3D datasets [6]. These challenges highlight the need for robust, automated segmentation solutions to process high-dimensional data with consistent accuracy.

Advanced segmentation methods for 3D medical imaging often involve deep learning models built with 3D convolutional neural networks [7]. The 3D U-Net model, with its encoder–decoder architecture,

is a model that is used for this purpose as it efficiently captures details across multiple scales, making it suitable for complex segmentation tasks [8]. A challenge in 3D image segmentation is the computational demand, which escalates with the size and complexity of the datasets [9,10]. Reducing computational time without compromising accuracy is imperative to make these tools practical for real-time clinical applications [11,12]. Enhanced segmentation algorithms that are both faster and more accurate could streamline clinical workflows, enabling expedited diagnosis and facilitating more rapid, data-driven decision-making [13].

Knowledge distillation is a compelling approach to addressing these computational challenges [14]. In this methodology, a smaller, more efficient model (the student) acquires knowledge from a high-performing, complex model (the teacher). This knowledge transfer preserves the teacher model's accuracy while reducing the complexity of the student

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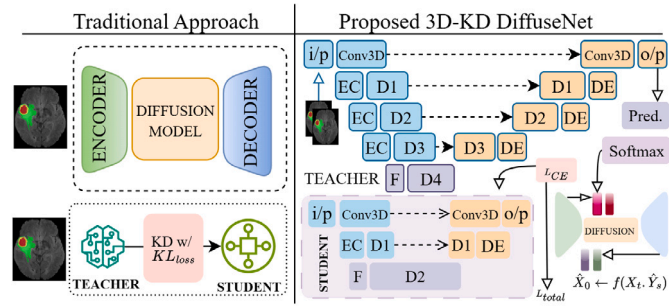


Fig. 1. Illustration of the limitations in conventional KD and diffusion models. Standard KD struggles to transfer robust representations, while diffusion models are computationally expensive. We address KD's representation limitations and diffusion models' high computational cost by integrating a diffusion-guided consistency loss that enhances feature learning without adding inference overhead.

model, making it faster and less computationally intensive [14,15]. This is particularly advantageous in medical imaging, where accuracy and efficiency are paramount. Knowledge distillation is a promising technique for tasks such as organ segmentation within large datasets.

SOTA studies [16–29] using knowledge distillation methodologies have certain limitations. The models often operate directly on raw 3D medical images without sufficient preprocessing, leading to an unnecessary computational burden. Denoising diffusion models have recently demonstrated remarkable performance in medical image segmentation due to their powerful nonlinear modeling and superior generalization capabilities [30]. These models iteratively refine noisy predictions toward a target distribution, effectively capturing structural details and mitigating boundary uncertainty. Despite these strengths, diffusion models remain relatively underutilized in the distillation process in the medical imaging domain.

Our approach addresses these shortcomings (see Fig. 1) by introducing a 3D Knowledge Distillation-based Diffusion Network (3DKD-DiffuseNet). The proposed framework reduces computational overhead while preserving segmentation accuracy. Our main contributions can be summarized as follows:

- We designed an intensity-based thresholding preprocessing and ROI identification step to localize relevant structures and optimize slice selection. This reduces computational demands while preserving critical spatial features.
- We introduced a novel teacher–student framework where a lightweight student model is trained using both soft predictions and denoising supervision via a forward–reverse diffusion process to enable robustness to input variability and structural ambiguity.
- A novel diffusion consistency loss is formulated that complements standard distillation objectives and enforces spatial fidelity through reconstruction to improve generalization across organs and modalities.
- We conduct extensive experiments on both brain and abdominal segmentation tasks across MRI and CT modalities. Our model achieves SOTA segmentation performance while significantly reducing inference time and computational cost.

2. Related works

Knowledge distillation-based methods. Recently, knowledge distillation (KD) has gained attention in medical image segmentation as a way to compress large models and improve generalization. For instance, in a work, Hu et al. [31] proposed KD-Net, where a mono-modal student is trained to mimic both the soft predictions and latent space of a multi-modal teacher. The teacher's weights are frozen, and the student

learns to match the teacher's bottleneck features. On BraTS2018, their T1ce-only student achieved 71.67% Dice for enhancing tumor (ET). Similarly, Rahimpour et al. [22] introduced a cross-modal distillation approach that transfers both soft predictions and intermediate features from a multi-sequence teacher to a single-sequence student. They tested variants with and without feature distillation and found the feature-level variant achieved 77.88% Dice for Whole Tumor using only T1w. In a related direction, Kang et al. [25] proposed MCTSeg, a CNN-Transformer hybrid framework that distills multimodal knowledge into unimodal student encoders. They included modules to transfer features, enhance unimodal signals, and simulate missing modalities during training. On BraTS2020, MCTSeg reached 87.24% Dice for Whole Tumor.

On the other hand, Dou et al. [32] addressed unpaired modality distillation between CT and MRI by sharing convolutional weights across modalities and aligning prediction distributions using KL divergence. Without needing registered data, their model improved the MRI segmentation Dice from 83.7% to 89.3% on abdominal organs. In another study, instead of relying on feature similarity, Vadachino et al. [33] used adversarial knowledge distillation in HAD-Net. In their approach, a student without access to T1ce learns from a full-modality teacher using hierarchical discriminators that align feature maps and outputs. On BraTS2019, HAD-Net achieved 73.3% Dice for enhancing tumor (ET). For weak supervision, Zhai et al. [34] proposed PA-Seg, which first expands sparse point annotations into dense pseudo-labels, then refines them using cross-knowledge distillation between two networks. Their method achieved 0.856 Dice on the brain tumor segmentation task. Unlike teacher–student setups, Yu et al. [35] introduced a self-distillation framework, ACW-SKD, which uses an auxiliary head and a feature reconstruction module to refine class-wise features within a single network. They combined this with a learnable temperature mechanism and achieved 73.74% and 83.18% Dice on Synapse and FLARE2022 datasets, respectively, using a compact ResNet18 backbone.

Most existing KD methods focus on transferring knowledge via soft labels or features, often assuming clean input data. While some handle missing modalities or weak labels, they do not explicitly train students to be robust against noisy inputs or structural ambiguity.

Diffusion-based models. Diffusion models, originally developed for generative image synthesis, have recently been adapted for medical image segmentation due to their ability to model complex structures and spatial uncertainty. For instance, in a work, Chen et al. [36] proposed HiDiff, a hybrid framework that combines a discriminative segmentor with a generative Bernoulli diffusion model (BBDM). They used an alternate-collaborative training strategy, including a binarized refiner and a cross transformer, and achieved 84.94% Dice on Synapse and 85.80% on BraTS2021. In another study, Chen et al. [37] introduced BerDiff, which employed Bernoulli noise as a discrete diffusion kernel tailored for binary segmentation. Unlike HiDiff, this model emphasized diversity in predictions through stochastic sampling and achieved 89.7% on BraTS2021. Similarly, Wolleb et al. [38] applied diffusion directly to segmentation masks using a DDPM-based approach. They used ground truth masks during training and generated multiple outputs for ensemble prediction, achieving 86.6% (single run) and 88.1% (ensemble) Dice on BraTS2020. Similarly, Xing et al. [39] proposed Diff-UNet, integrating diffusion within a U-Net architecture. They introduced a Denoising module and a Step-Uncertainty Fusion mechanism, reporting 85.35% on BraTS2020 and 73.69% on MSD Liver.

Some works used diffusion more indirectly for refinement or enhancement. For example, Li et al. [40] proposed CorrDiff+, a corrective model that refines initial segmentations by addressing systematic errors using a VQ-VAE and multi-fusion attention. It achieved 87.48% Dice on BraTS2019 and 93.02% on the Jun Cheng dataset. In another work, Zhang et al. [41] introduced DiffBoost, which uses diffusion models for data augmentation rather than direct segmentation. By generating synthetic images with text and edge guidance, it achieved 84.56% on

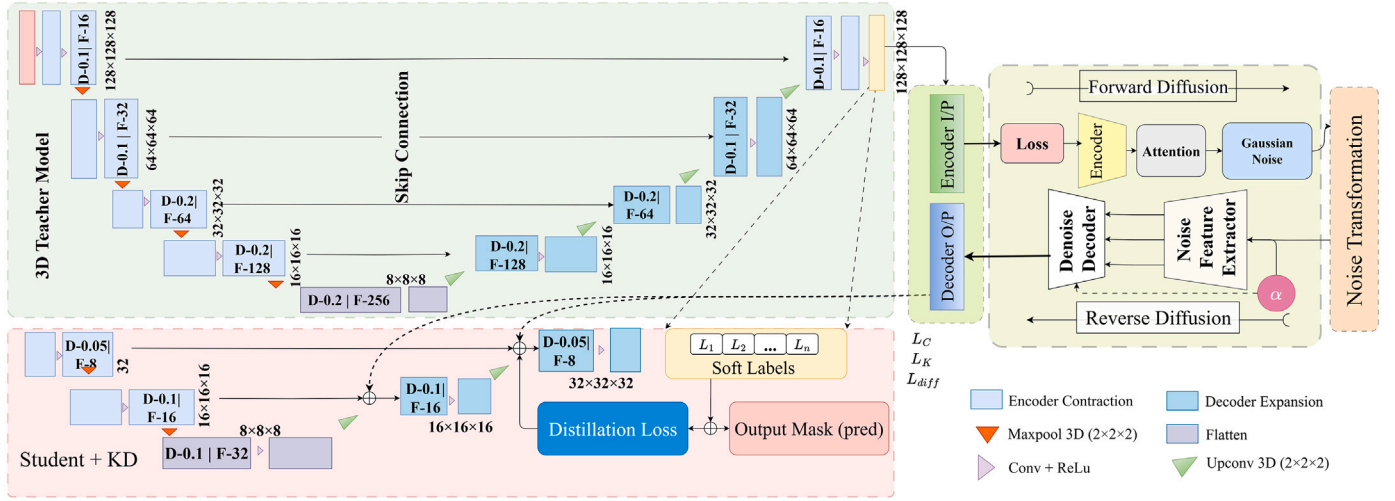


Fig. 2. Overview of the proposed 3DKD-DiffuseNet with diffusion-based KD and attention mechanisms. The teacher model is a 3D attention-based encoder–decoder that extracts rich feature representations, which are transferred to the lightweight student model. The student model incorporates a diffusion process to enhance feature learning, with segmentation and distillation losses used to refine its predictions.

Prostate MRI, 94.78% on Spleen CT, and 71.65% on Breast Ultrasound. More recently, Wu et al. proposed MedSegDiff [42] and MedSegDiff-V2 [43], which incorporate conditioning strategies to guide the diffusion process more effectively. MedSegDiff used Dynamic Conditional Encoding and a frequency parser, while MedSegDiff-V2 introduced spatial and spectral transformers. These models achieved 88.9–89.9% and 90.8% Dice on brain tumor datasets, respectively.

Despite notable progress, diffusion-based segmentation models share common limitations. Most models rely on full denoising inference, which is computationally expensive and slow, making real-time deployment difficult. Even with optimizations like binarization or VQ-VAE, these models remain resource-intensive. Corrective models are dependent on the quality of initial segmentations, and augmentation-based methods may introduce bias or lack structural variability. Although few studies [44–47] have attempted to fuse knowledge distillation with diffusion, the implications are notably absent in 3D medical image segmentation tasks. Our work addresses these limitations by introducing a lightweight diffusion-enhanced student model trained within a knowledge distillation framework, where diffusion is used not for generation but to improve robustness and generalization without compromising speed.

3. Methods

In this section, we describe the preprocessing steps for segmenting anatomical regions in the datasets, followed by the 3DKD-DiffuseNet model. This model leverages a teacher–student framework with diffusion-based knowledge distillation and attention mechanisms to improve segmentation accuracy. The architecture of our proposed teacher–student model with diffusion, 3DKD-DiffuseNet, is depicted in Fig. 2.

3.1. Dataset details

We evaluated our method on two distinct 3D medical image segmentation tasks to demonstrate its broad applicability: (1) brain tumor segmentation using the BraTS2017, BraTS2018, and BraTS2019 datasets [48–50], with additional generalizability validation on BraTS2020 and BraTS2021; and (2) abdominal organ segmentation using the RAOS dataset [51,52].

BraTS benchmark. The BraTS benchmark datasets include patients diagnosed with high-grade glioblastomas and low-grade glioblastomas. Each patient scan consists of four MRI modalities: T1-weighted, post-contrast T1-weighted (T1ce), T2-weighted, and Fluid Attenuated

Inversion Recovery (FLAIR). In the dataset annotations, every voxel is assigned to one of the raw labels: Label 1 corresponds to the necrotic or non-enhancing tumor core (NCR), Label 2 to edema (ED), and Label 4 to enhancing tumor (ET). Following the standard segmentation evaluation metrics, these raw labels are then combined into the three evaluated subregions used in our study: whole tumor ($WT = 1 \cup 2 \cup 4$), tumor core ($TC = 1 \cup 4$), and enhancing tumor ($ET = 4$). Background voxels are labeled separately (Labeled as 0). The results reported in the subsequent sections follow these benchmark guidelines [53].

For the segmentation task, we used the officially designated test sets and evaluated performance using the same offline evaluation methodology applied in the original challenges.

RAOS. The RAOS dataset comprises 413 clinical computed tomography (CT) scans and 3,717 magnetic resonance (MR) scans, with nine sequences associated with each case. Annotations cover 19 abdominal organs, each verified by an experienced oncologist. The dataset is split into three subsets to represent different surgical contexts: cases without surgery (Set A, 317), cases after surgery without organ removal (Set B, 22), and cases after surgery with one or more organs missing (Set C, 74). Following the official RAOS study, we used Set A for model development (220 for training and the rest for validation). Since Sets B and C contained fewer samples, we also used them only for testing, as per the official implementation by the authors. This dataset presents challenging cases not covered in other public datasets, such as prostate and seminal vesicles.

3.2. 3D-image preprocessing

The preprocessing stage involves isolating key anatomical regions for both the BraTS and the RAOS datasets. Using intensity-based thresholding, brain regions were accurately segmented in 3D MRI volumes for the BraTS benchmark datasets. Similarly, specific abdominal organs (e.g., liver, kidneys, spleen) were isolated in 3D CT volumes for the RAOS dataset by tailoring intensity thresholds to match each organ's characteristics. To mitigate variability arising from differing acquisition parameters across MRI sequences, voxel intensities were standardized using Eq. (1):

$$I_{\text{standardized}} = \frac{I - \mu}{\sigma} \quad (1)$$

where I is the raw voxel intensity, μ is the mean intensity of the sequence, and σ is the standard deviation of the sequence. This transformation ensures that the voxel intensities have a zero mean and a unit

variance, improving consistency across samples. In 3D CT volumes for the RAOS dataset, to ensure uniformity across contrast-enhanced and non-contrast CT scans, voxel intensities are normalized to identify suitable intensity thresholds for each organ. Regions of interest (ROIs) were then identified using segmentation masks, rather than fixed cropping, to ensure that only relevant anatomical structures are included while irrelevant areas are discarded. This is defined in Eq. (2):

$$M(x, y, z) = \begin{cases} 1 & \text{if } (x, y, z) \in \text{ROI} \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

where M denotes the segmentation mask function. This approach preserves critical anatomical features and ensures the model remains anatomy-aware, while reducing computational burden. Following this, the slice range containing relevant data was determined using Eq. (3):

$$\begin{aligned} s_{\min} &= \min\{s \mid M(:, :, s) \neq 0\}, \\ s_{\max} &= \max\{s \mid M(:, :, s) \neq 0\} \end{aligned} \quad (3)$$

With this step, we reduce unnecessary training overhead without compromising segmentation accuracy by excluding slices that do not contain relevant structures (e.g., no-tumor or no-organ slices). Finally, to manage large data volumes and enable faster training, a custom data generator was implemented to compile 3D volumes into arrays. The compiled batch B consists of n preprocessed 3D volumes, where each volume is denoted by V_i , with i ranging from 1 to n .

3.3. 3DKD-DiffuseNet

Unlike traditional KD approaches that assume architectural similarity, our framework is agnostic to the internal structure of the student model. As long as the student can produce output logits compatible with the teacher (e.g., same number of classes), the architecture can vary in terms of depth, width, and presence of auxiliary modules. Algorithm 1 provides a brief overview of the proposed methodology. Further architectural details are provided in the following subsections.

Algorithm 1 3DKD-DiffuseNet: Diffusion-Based Knowledge Distillation for Multi-Organ Segmentation

```

1: Input: training img  $X$ , GT label  $Y$ 
2: // train teacher model
3: initialize 3D Attention U-Net  $f_T$ 
4: for each training epoch do
5:   for each batch  $(X, Y)$  do
6:     predict labels:  $\hat{Y}_T = f_T(X)$ 
7:     compute teacher loss:  $L_T$ 
8:     update  $f_T$  using gradient descent
9:   end for
10: end for
11: // train student with distillation and diffusion
12: initialize lightweight student model  $f_S$ 
13: for each training epoch do
14:   for each batch  $(X, Y)$  do
15:     predict teacher soft labels:  $\hat{Y}_T = f_T(X)$ 
16:     predict student labels:  $\hat{Y}_S \leftarrow f_S(X)$ 
17:     compute CE loss:  $L_C \leftarrow -\sum Y \log(\hat{Y}_S)$ 
18:     compute distillation loss:  $L_K \leftarrow \sum \hat{Y}_T \log\left(\frac{\hat{Y}_T}{\hat{Y}_S}\right)$ 
19:     add Gaussian noise:  $X_t \leftarrow X + \mathcal{N}(0, \sigma_t^2)$ 
20:     denoise input:  $\hat{X}_0 \leftarrow f_{\text{denoise}}(X_t, \hat{Y}_S)$ 
21:     compute diffusion loss:  $L_{\text{diff}} \leftarrow \|\hat{X}_0 - X\|_2^2$ 
22:     combine:  $L_{\text{total}} \leftarrow \alpha L_C + \beta L_K + \gamma L_{\text{diff}}$ 
23:     update  $f_S$  using gradient descent
24:   end for
25: end for
26: return trained diffusion-based student model  $f_S$ 

```

3.3.1. Teacher model

The teacher model is a 3D attention-based encoder–decoder model built to segment volumetric medical images. The teacher model is visualized in Fig. 3.

The model follows a contraction–expansion path to the downsample and then the upsample of the input. The input volumes enter through layers with a $3 \times 3 \times 3$ convolution, 16 filters, and ReLU activation. Max-pooling reduces the spatial dimensions in each layer, progressively doubling the number of filters (16, 32, 64, 128, 256) until reaching the bottleneck, which captures detailed spatial features. Dropout is applied at increasing rates to promote generalization, especially in the bottleneck layer, where it is set to 0.3. For the 3D convolution operation, the input preprocessed image can be denoted as $X \in \mathbb{R}^{H \times W \times D \times C}$, where H , W , and D represent the height, width, and depth of the image, and C is the number of channels. The feature map output at layer i is represented by Eq. (4):

$$F_i = \sum_{c=1}^{C_{i-1}} W_i^c * X_{i-1}^c + b_i \quad (4)$$

where W_i^c is the 3D convolution kernel applied to the c th channel at layer i , and b_i is the bias term. Following this, a 3D convolution is applied to this input, followed by batch normalization (BN) and ReLU activation, σ . A general 3D convolution applied to an input tensor X with a kernel W can be expressed by Eq. (5):

$$F(h, w, d, c) = \sum_{h', w', d', c'} W(h', w', d', c') \cdot X(h - h', w - w', d - d', c') \quad (5)$$

where the convolution is computed over the height h , width w , and depth d of the input. The convolutional kernel has dimensions of $3 \times 3 \times 3$, and the number of output channels increases with depth, often doubling after each down-sampling step. For our teacher model, to reduce the spatial dimensions, MaxPooling3D selects the maximum value within non-overlapping windows of size $2 \times 2 \times 2$, to reduce the spatial dimensions. For a feature map $F_i \in \mathbb{R}^{H_i \times W_i \times D_i \times C_i}$, the output feature map after the pooling operation is described by Eq. (6):

$$F_{po,i}(h, w, d, c) = \max_{(h', w', d')} (F_i(h', w', d', c)) \quad (6)$$

where h' , w' , and d' are spatial coordinates within each pooling window. The dimensions are reduced, halving H_i , W_i , and D_i . The general formula for representing the MaxPooling is given by Eq. (7), where h' , w' , and d' represent local positions within the window.

$$F_{po}(h, w, d, c) = \max_{(h', w', d') \in [0, s)} F(h + h', w + w', d + d', c) \quad (7)$$

We have incorporated the attention mechanism to enhance the model's focus on important regions. An attention map A_i is computed using a 3D convolution followed by a sigmoid activation. Here, the A_i is an attention mask in the range $[0, 1]$ which highlights important spatial regions in the feature map F_i . The output feature map, denoted as $F_{at,i}$, is then obtained by element-wise multiplication of the attention map with the original feature map, which ensures that relevant features are enhanced while less important ones are suppressed. Following this, the decoder uses 3D transposed convolutions to upsample the feature maps, restoring spatial dimensions. Given the feature map $F_{at,i+1}$ from a deeper layer, the upsampled feature map is computed as the transposed convolution operation followed by an activation.

We then added skip connections between the corresponding encoder and decoder layers to allow the model to retain spatial details. Following this, upsampled feature maps with their corresponding encoder outputs are concatenated. The final layer of the Teacher model applies the SoftMax function to generate voxel-wise probabilities. For a voxel v and class k , the SoftMax output is given by Eq. (8):

$$\hat{Y}_T(v, k) = \frac{\exp(Z_T(v, k))}{\sum_{j=1}^K \exp(Z_T(v, j))} \quad (8)$$

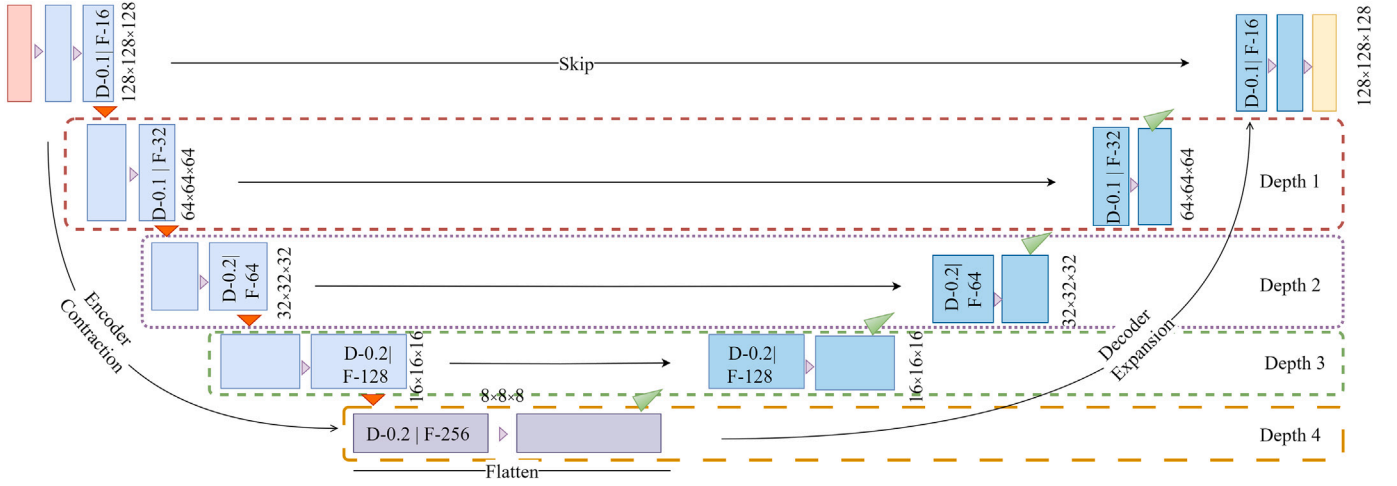


Fig. 3. Architecture of the teacher model, a standard 3D attention U-Net with encoder-decoder structure. It employs 4-depth downsampling and upsampling paths, attention mechanisms for focusing on relevant regions, and skip connections to preserve spatial details. The final output generates voxel-wise segmentation probabilities via SoftMax.

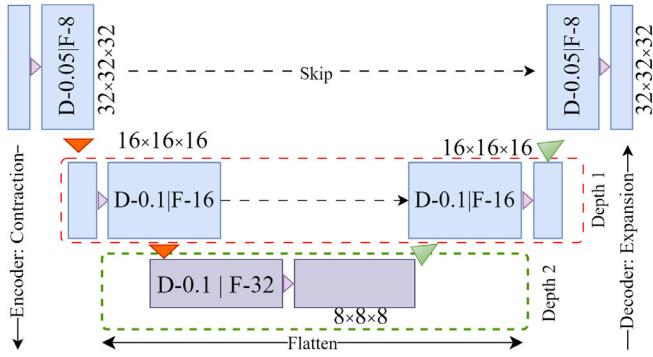


Fig. 4. Visualization of the student model, designed as a lightweight variant of the teacher. It uses a 2-depth architecture with fewer layers and filters to reduce computational cost. Despite being compact, it benefits from both teacher supervision (via KD) and the diffusion-based regularization introduced in our framework.

where $Z_T(v, k)$ is the logit for class k and K is the total number of classes. After obtaining the optimal weight from the teacher model, a student model that distills knowledge from the teacher model is implemented.

3.3.2. Student model

The student model is a lighter variant of the teacher model, designed with fewer layers, fewer filters, and no attention blocks. It is optimized for reduced computational load while maintaining competitive segmentation performance. The model architecture is shown in Fig. 4.

After training the teacher, the student model is supervised through both the teacher's soft-label outputs and the original ground truth annotations. Initially, the student model learns to align its predictions with the teacher's output distribution using Kullback-Leibler (KL) divergence. The softmax outputs of the teacher and student models at voxel v for class k are denoted by $\hat{Y}_T(v, k)$ and $\hat{Y}_S(v, k)$, respectively. The distillation loss is computed as shown in Eq. (9):

$$L_K = \sum_v \sum_{k=1}^K \hat{Y}_T(v, k) \log \left(\frac{\hat{Y}_T(v, k)}{\hat{Y}_S(v, k)} \right) \quad (9)$$

Additionally, the student is trained using the standard cross-entropy (CE) loss with ground truth labels $Y(v, k)$. We calculate the CE loss L_C

following Equation (10):

$$L_C = - \sum_v \sum_{k=1}^K Y(v, k) \log (\hat{Y}_S(v, k)) \quad (10)$$

The KL loss L_K (see Eq. (9)) transfers the teacher's complex predictive distribution to the student, but it operates only on the final output logits. This provides limited guidance on feature representation formation for noisy inputs. Thus, we introduce a complementary diffusion-based objective, which forces the student's encoder to learn a fundamental, noise-invariant understanding of the input structure by reconstructing clean volumes from noisy versions. Consequently, the student then more effectively aligns its output with the teacher's soft labels (L_K), as its improved features are more capable of capturing the necessary semantics. The diffusion loss (detailed in Section 3.3.3) thus works synergistically with L_K : one improves feature robustness, while the other refines the final output based on those improved features.

3.3.3. Student diffusion

Forward and reverse diffusion. Unlike generative diffusion models that perform iterative denoising during inference, we repurpose diffusion as a regularizer to enhance the quality of the student model's feature representations. To simulate model robustness, we incorporate a forward and reverse diffusion process during training. In the forward diffusion step, Gaussian noise is progressively added to clean inputs according to a noise schedule controlled by variance σ_t . This noisy input, denoted as X_t , is formulated as shown in Eq. (11):

$$X_t = X + N(0, \sigma_t^2) \quad (11)$$

In the reverse diffusion step, the student model's feature encoder itself acts as the denoising function to estimate the original image from the noisy input. This process teaches the encoder to learn noise-invariant features and iteratively removes noise, guided by the model's predictions. The denoised output $\hat{X}_0$ is computed as shown in Eq. (12):

$$\hat{X}_0 = f_{\text{denoise}}(X_t, \hat{Y}_S) \quad (12)$$

Diffusion-augmented knowledge distillation. Standard knowledge distillation transfers the teacher's knowledge through the alignment of output distributions between the teacher and student. This process is typically realized by minimizing the Kullback-Leibler divergence between the softmax predictions of the two networks. However, this formulation supervises only the final prediction layer of the student

Table 1

Dice score (%) comparison of teacher and student model on the BraTS benchmark datasets (2017–2019) across 4 modalities: T1-weighted (T1), Enhanced T1-weighted (T1-Enh), T2-weighted (T2), and Flair. Subregions include enhancing tumor (ET), tumor core (TC), and whole tumor (WT). Results are reported separately for high-grade (HG) and low-grade (LG) gliomas.

Grade	Dataset	Subregion	Teacher Model Dice Score				Student Model Dice Score			
			T1	T1-Enh	T2	Flair	T1	T1-Enh	T2	Flair
High Grade (HG)	BraTS2017	ET	91.5	91.1	91.1	92.7	94.8	95.1	94.7	95.1
		TC	89.6	89.2	89.2	90.5	92.6	93.5	92.9	93.5
		WT	92.2	91.1	89.6	91.7	95.1	94.4	94.1	95.6
		Avg.	91.1	90.5	89.97	91.7	94.2	94.3	93.9	94.7
	BraTS2018	ET	90.2	90.2	91.1	91.8	93.3	93.9	94.2	94.5
		TC	89.3	89.9	89.6	90.7	92.9	93.2	92.6	93.9
		WT	90.3	90.3	90.3	91.6	94.6	93.8	94.5	94.2
		Avg.	89.93	90.14	90.33	91.37	93.6	93.6	93.7	94.2
	BraTS2019	ET	89.8	89.4	90.5	93.4	92.7	93.0	93.7	95.3
		TC	88.9	88.9	90.1	90.7	92.1	92.2	93.0	94.2
		WT	90.4	90.5	90.3	92.6	93.4	94.1	94.1	95.1
		Avg.	89.7	89.6	90.3	92.23	92.7	93.1	94.33	95.1
Low Grade (LG)	BraTS2017	ET	89.8	89.8	90.1	89.9	92.8	93.0	93.2	93.9
		TC	86.7	86.7	88.9	90.0	89.9	91.7	92.1	92.1
		WT	89.3	89.3	90.3	90.5	93.1	93.9	93.9	94.3
		Avg.	88.6	88.60	87.77	90.14	91.9	92.8	93.0	93.2
	BraTS2018	ET	89.8	89.8	90.2	90.1	92.9	92.9	93.4	92.9
		TC	87.4	88.2	88.4	89.6	90.8	91.3	91.3	91.5
		WT	88.5	89.9	90.2	90.4	92.5	93.7	93.6	93.1
		Avg.	87.47	89.3	89.60	90.03	92.0	92.6	92.7	92.5
	BraTS2019	ET	88.2	89.1	89.8	88.7	91.4	92.2	92.9	92.4
		TC	86.1	87.5	88.2	87.1	89.7	90.7	91.0	89.9
		WT	89.1	89.2	90.1	89.5	92.1	92.9	93.2	92.1
		Avg.	87.8	88.6	89.37	84.3	91.0	92.0	93.2	92.1

and does not provide direct constraints on the intermediate feature representations. As a result, the student can reproduce the teacher's logits but still learn suboptimal or unstable features (see empirical findings in Section 4.3), especially around ambiguous tumor boundaries where the teacher itself may be uncertain.

In this framework, the student is exposed to noisy inputs during training (via forward diffusion) and learns to reconstruct clean input representations through its own representations through a denoising function (reverse diffusion). This process introduces a novel auxiliary diffusion consistency loss L_{diff} that enforces accurate reconstruction and effectively regularizes the student's feature extractor. We define the loss in Eq. (13):

$$L_{\text{diff}} = \|\hat{X}_0 - X\|_2^2 \quad (13)$$

This loss complements the standard CE (Eq. (10)) and KL terms (Eq. (9)) by guiding the model to produce representations invariant to noise. The entire diffusion process is used only during training and is discarded during inference. Thus, it ensures no computational overhead in the student model. The final loss function for training the student integrates all three components. It is defined in Eq. (14):

$$L_{\text{total}} = \alpha L_C + \beta L_K + \gamma L_{\text{diff}} \quad (14)$$

where α , β , and γ are hyperparameters that balance the contribution of the segmentation loss, distillation loss, and diffusion consistency loss, respectively.

4. Experimental results

4.1. Evaluation criteria

Our proposed methodology is evaluated using a range of performance metrics. These include Accuracy (Acc), Precision (Prec), Sensitivity (Sen), and Specificity (Spe). Additionally, we used the Dice Score to calculate the overlap between the predicted and true segmentation.

It is worth noting that the **t/** and **s/** suffixes preceding any metrics represent **Teacher** and **Student**, respectively, in Tables 7, 8, and 9. For all SOTA comparison results reported in this study, the values were directly sourced either from the respective publications or from the subsequent benchmark studies to avoid implementation bias.

4.2. Training analysis

All experiments were carried out on a system equipped with an AMD Ryzen 5 5600X 6-core CPU and 16 GB of RAM, along with a ZOTAC GAMING GeForce RTX 3060 Twin Edge OC GPU that has 12 GB of VRAM. Our training configuration involved running the teacher model and the student model, which incorporates diffusion, for 100 epochs (each). For both the teacher and the student models, we used the Adam optimizer with a learning rate of 0.001, and the batch size was set to 2. The choice of batch size 2 is due to the high computational cost of training on large 3D volume imaging datasets [54,55]. During training, both models achieved over 99% accuracy, with validation accuracy consistently above 98%.

4.3. Diffusion impact on the teacher–student distillation

Direct teacher-to-student transfer supervises the student primarily through the alignment of output distributions. While this strategy passes semantic information from the teacher to the student, it does not constrain how the student forms intermediate feature representations. In practice, this can also lead to unstable or noisy features in the student encoder. As a result, the benefit of direct transfer is often limited to small improvements in prediction accuracy. On the other hand, the student model, incorporating diffusion, can potentially outperform the teacher model across imaging types (see Table 1). To verify this effect, we trained the teacher and student models for 50 epochs (each) on a subset of the BraTS2019 dataset. Dice scores averaged across four MRI modalities are reported in Table 2, where we compared (i) direct teacher-to-student transfer without diffusion and (ii) teacher-to-student transfer with diffusion.

Although it is observed that direct transfer increases Dice scores, the performance increment is limited to a slight improvement compared to the teacher. However, the student achieves clear improvements across all subregions with the diffusion integration, with an average Dice gain of nearly 4%. These results demonstrate that diffusion provides stronger and more stable supervision than direct teacher-to-student transfer. The following sections report student performances corresponding to the diffusion-augmented variant.

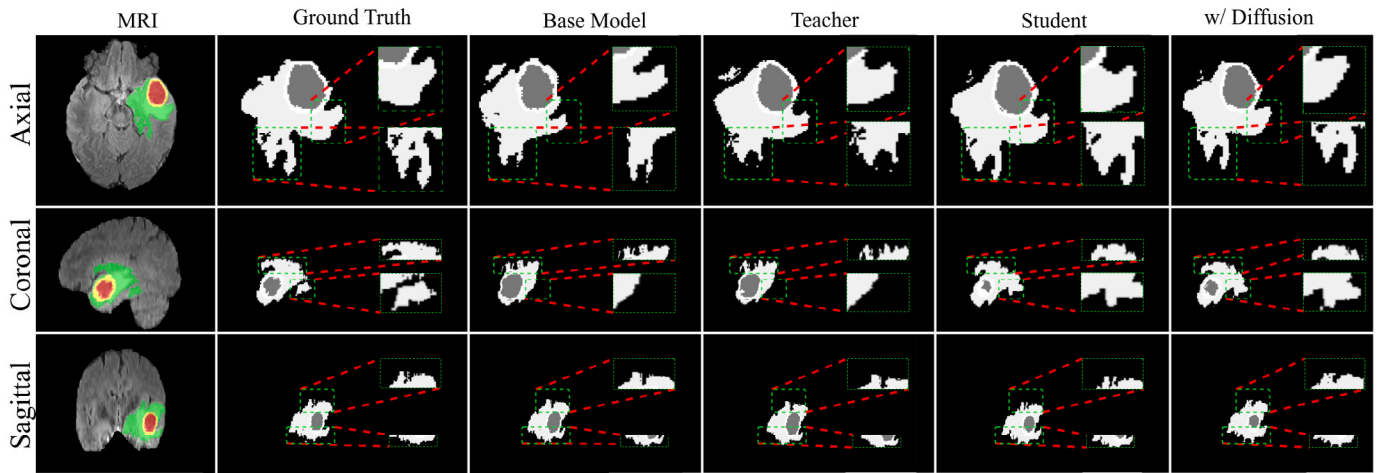


Fig. 5. Qualitative segmentation results on the BraTS benchmark datasets. The local ground truth annotations and the tumor segmentation outputs are highlighted and zoomed in for each approach to show the segmentation outline of the tumor subregions. Images in the last column (i.e., w/Diffusion) are the segmentation visualization achieved with the (final) diffusion-based student model.

Table 2

Comparison of teacher-to-student transfer with and without diffusion on BraTS2019 (average across four modalities). Dice scores (%) are reported for enhancing tumor (ET), tumor core (TC), and whole tumor (WT).

Subregion	Dice (%)		
	Teacher	Std w/o Diff	Std w/ Diff
ET	74.82	76.01	80.33
TC	72.47	73.89	76.20
WT	77.31	77.93	81.66
Avg.	74.87	75.94	79.40

4.4. Evaluation on the BraTS benchmark datasets

As seen in Table 1, the proposed student model, incorporating diffusion, consistently outperformed the teacher model across all datasets, tumor grades, and imaging types. For high-grade (HG) tumors, the student model achieved Dice scores between 0.942 and 0.947 across T1-weighted, Enhanced T1-weighted, T2-weighted, and Flair images, while the teacher model scored between 0.899 and 0.917. Notably, on Flair images of HG tumors, the student model reached a peak Dice score of 0.956, compared to the teacher's 0.927. Table 3 shows the detailed results for high-grade tumor classification.

For low-grade (LG) tumors, the student model's results were on average 3%–5% higher than the teacher model, with scores ranging from 0.910 to 0.932 across all modalities. It performed best on BraTS2017 HG tumors with an average Dice of 94.27%; and also showed strong stability across all datasets. Flair images generally yielded the highest accuracy for both models.

On both HG and LG segmentation tasks, the proposed student model achieved average Dice of 93.50%, 93.11%, and 92.93% on the BraTS2017, 2018, and 2019 datasets, respectively. Fig. 5 illustrates the segmentation output.

Our student model demonstrated remarkable performance in classifying high-grade tumors across the BraTS benchmark datasets as well. For instance, on BraTS2017, we achieved an accuracy of 0.985 for enhancing tumors (ET) using T1-weighted images, with a precision of 0.976 and a sensitivity of 0.980. For tumor core (TC), the accuracy was 0.978, and we reached an accuracy of 0.990 for the whole tumor (WT) classification.

On the BraTS2018 dataset, our model maintained a high performance, scoring an accuracy of 0.987 for ET in T1-weighted images, 0.981 for Enhanced T1-weighted images, and 0.993 for WT classification. The results were consistent across the tumor grades. Notably, in

the high-grade tumors, we achieved average Dice scores of 94.27%, 93.77%, and 93.80% on the BraTS2017, 2018, and 2019 datasets, respectively. Similarly, in the low-grade tumor segmentation tasks, we scored Dice scores of 92.72%, 92.45%, and 92.07% on the respective datasets.

The student model demonstrated a strong performance in low-grade tumor evaluations across the BraTS benchmark datasets as well (see Table 4). For instance, the model achieved an accuracy of 0.967 for ET in T1-weighted images in BraTS2017. For TC classification, the accuracy increased to 0.972, and for WT, it further improved to 0.978. Similarly, for BraTS2018, the performance remained consistent. The model achieved an accuracy of 0.964 for ET and demonstrated remarkable results for WT classification, with a peak accuracy of 0.999 in T2-weighted images. It also performed well for TC classification, reaching 0.975. The results from BraTS2019 continued to showcase the model's reliability. In contrast to the HG tumor segmentation task, there was a lower performance for low-grade tumors, with a maximum accuracy of 97.8% for enhancing tumors in T1-weighted images in BraTS2017. High-grade tumors also exhibited higher precision and sensitivity across various modalities. For example, the precision for whole tumor classification reached 0.998 for high-grade tumors.

While comparing our model with SOTA KD models, we also notice this improvement across multiple datasets. Existing KD methods follow a standard approach, where a student network learns from a teacher trained on multimodal MRI data. For instance, Cross-Modal KD [22] and DWKD [24] transferred knowledge to unimodal students using feature- and pixel-level distillation and achieved Dice scores up to 75.20% and 88.66% for TC and WT segmentation, respectively. Similarly, MCTSeg [25], POD [26], and TASCNet [27] leverage multi-scale and collaborative strategies, with Dice scores ranging from 57.44% to 84.30% across tumor subregions. On the other hand, MKD [28], MTCM-KD [29], KD-Net [31], HAD-Net [33], and PA-Seg (SCM) [34] use multi-teacher, adversarial, and pseudo-labeling schemes and achieve Dice scores between 66.70% and 87.90% for different tumor subregions.

Compared to these studies, our model consistently achieves higher Dice scores across BraTS2017–2019. For example, on BraTS2017 high-grade tumors, it reaches 94.92%, 93.12%, and 94.80% for ET, TC, and WT, respectively, and outperforms all the listed KD models. Similar improvements are observed for low-grade tumors and on other datasets. Fig. 6 shows the comparative segmentation overview of our proposed model with several SOTA models; and the comparative result of our model against SOTA segmentation models on the BraTS2017-19 datasets is given in Table 5.

Table 3

High grade (HG) tumor classification performance of student model on the BraTS benchmark datasets (2017–2019). Metrics include Accuracy (Acc), Precision (Prec), Sensitivity (Sen), and Specificity (Spe), evaluated across four MRI modalities (T1, T1ce, T2, FLAIR). Results are reported in %.

Dataset	Subregion	T1-weighted				Enhanced T1-weighted				T2-weighted				Flair			
		Acc	Prec	Sen	Spe	Acc	Prec	Sen	Spe	Acc	Prec	Sen	Spe	Acc	Prec	Sen	Spe
BraTS2017	ET	98.5	97.6	98.0	98.8	97.4	98.1	97.8	98.5	98.5	98.9	98.7	98.8	98.5	99.0	98.8	99.2
	TC	97.8	97.9	97.5	98.1	98.6	97.9	97.5	99.0	99.2	99.0	99.1	99.3	99.4	99.1	99.3	99.5
	WT	99.0	98.2	98.8	99.1	99.1	98.8	99.3	99.6	99.8	99.5	99.6	99.7	99.8	99.6	99.7	99.9
	Avg.	98.4	97.9	98.1	98.7	98.4	98.3	98.2	99.0	99.2	99.1	99.1	99.3	99.2	99.2	99.3	99.5
BraTS2018	ET	98.7	97.9	98.5	98.9	97.8	97.7	97.9	98.2	98.6	98.1	98.4	98.8	98.1	98.4	98.6	98.8
	TC	98.1	97.5	97.3	97.9	99.5	99.4	99.1	99.3	99.4	99.3	99.5	99.6	99.6	99.3	99.1	99.4
	WT	99.3	99.1	99.6	99.4	98.8	97.5	97.6	98.1	99.1	99.4	99.2	99.0	98.7	98.5	98.6	98.8
	Avg.	98.7	98.2	98.7	98.7	98.7	98.2	98.2	98.5	99.0	98.9	99.0	99.1	98.8	98.7	98.8	99.0
BraTS2019	ET	99.6	99.3	99.4	99.5	98.9	99.1	98.8	98.7	99.9	99.7	99.8	99.9	99.9	99.8	99.6	99.7
	TC	98.2	98.1	97.9	98.3	98.2	98.1	98.4	98.7	98.7	99.1	98.8	98.9	98.2	98.1	98.4	98.9
	WT	99.8	98.6	99.1	99.2	99.6	99.4	99.2	99.5	99.3	99.6	99.4	99.5	99.6	99.5	99.8	99.1
	Avg.	99.2	98.7	98.8	99.0	98.9	98.9	98.8	99.0	99.3	99.5	99.3	99.4	99.2	99.1	99.3	99.2

Table 4

Classification performance of the student model on low-grade (LG) tumors from the BraTS benchmark datasets (2017–2019). Metrics include Accuracy (Acc), Precision (Prec), Sensitivity (Sen), and Specificity (Spe), evaluated across four MRI modalities. Results are reported in %.

Dataset	Subregion	T1-weighted				Enhanced T1-weighted				T2-weighted				Flair			
		Acc	Prec	Sen	Spe	Acc	Prec	Sen	Spe	Acc	Prec	Sen	Spe	Acc	Prec	Sen	Spe
BraTS2017	ET	96.7	96.5	96.6	96.8	96.5	97.0	96.8	96.6	96.7	97.2	97.5	96.8	97.5	97.8	97.6	97.7
	TC	97.2	97.8	97.1	97.3	97.2	96.8	97.1	97.3	98.1	97.8	98.5	98.4	98.9	98.5	98.8	99.0
	WT	97.8	97.5	97.7	97.9	98.8	98.6	98.7	98.9	99.5	99.3	99.4	99.6	99.6	99.4	99.5	99.7
	Avg.	97.2	97.3	97.1	97.3	97.5	97.5	97.5	97.6	98.1	98.1	98.5	98.3	98.7	98.6	98.6	98.8
BraTS2018	ET	96.4	96.2	96.5	96.6	97.4	97.5	97.3	97.6	97.4	97.6	97.3	97.5	98.2	97.9	98.1	98.3
	TC	97.5	97.3	97.4	97.6	96.0	96.2	96.4	96.1	96.6	96.5	96.2	96.4	97.1	97.3	97.5	97.4
	WT	96.1	96.0	96.2	96.3	99.0	98.9	99.1	99.2	99.9	99.8	99.7	99.9	99.8	99.7	99.5	99.9
	Avg.	96.7	96.5	96.7	96.8	97.5	97.5	97.6	97.6	98.0	98.0	97.7	97.9	98.4	98.3	98.4	98.5
BraTS2019	ET	97.6	97.4	97.5	97.7	96.7	96.5	96.6	96.8	98.6	98.4	98.5	98.7	97.8	98.1	98.4	97.9
	TC	96.8	96.6	96.7	96.9	98.1	98.2	97.9	98.2	97.9	96.8	97.2	97.1	97.4	97.6	97.3	97.8
	WT	97.9	96.8	97.1	97.2	98.3	98.5	98.4	98.5	99.9	99.1	98.9	99.2	99.2	99.1	99.3	99.4
	Avg.	97.4	96.9	97.1	97.3	97.7	97.7	97.6	97.8	98.8	98.1	98.2	98.3	98.1	98.3	98.3	98.4

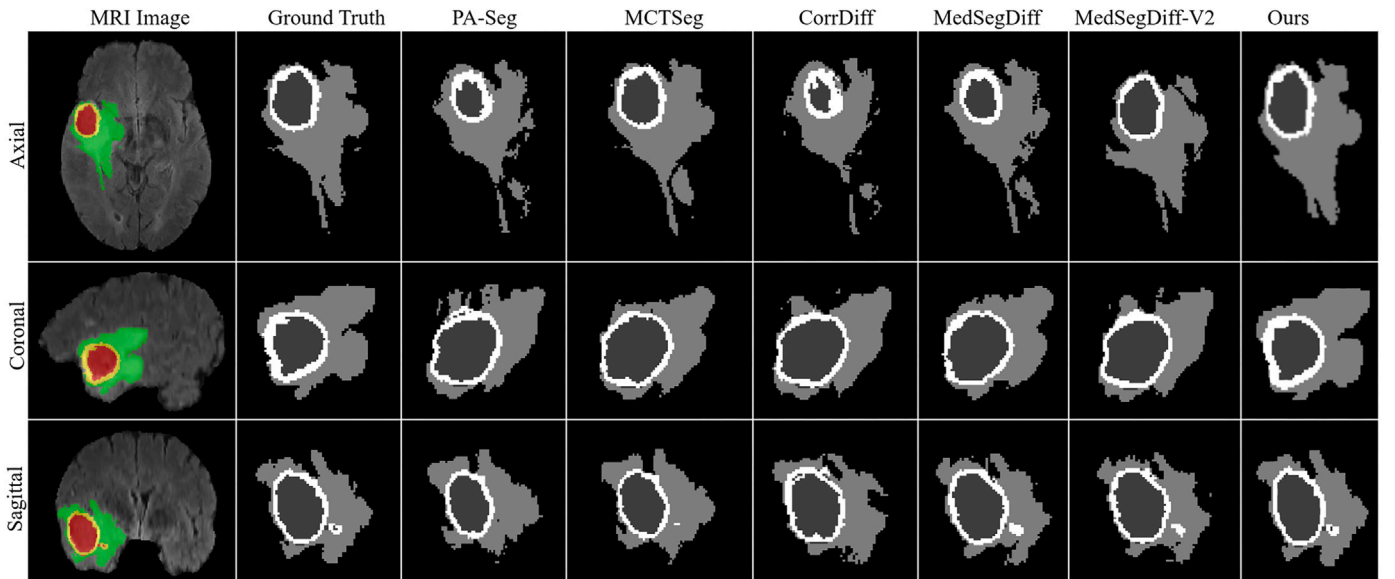


Fig. 6. Comparison with SOTA segmentation models on the BraTS benchmark dataset. The second column shows the ground truth masks, while the remaining columns display the predicted segmentation masks of different models compared against the ground truth. Results from our diffusion-based KD student model are shown in the final column.

We evaluated our model on both BraTS2020 and BraTS2021 datasets alongside SOTA Diffusion-based segmentation models to assess generalizability. On BraTS2020, 3DKD-DiffuseNet achieved a Dice score of 92.01%, which exceeding the next-best model (DDPM ensemble [38],

90.9%) by +1.11% points. On BraTS2021, our model reached 90.57%; it closely matches the top-performing MedSegDiff-V2 [43] (90.8%), with a difference of only 0.23% points. These results demonstrate that 3DKD-DiffuseNet generalizes well across datasets while maintaining

Table 5

Comparison of segmentation performance (Dice score in %) of the proposed 3DKD-DiffuseNet against SOTA KD models on the BraTS benchmark datasets (2017–2019). Results are reported for enhancing tumor (ET), tumor core (TC), and whole tumor (WT) across both HG and LG gliomas.

Model	Dataset	ET	TC	WT
Cross-Modal KD [22]	BraTS2018	–	71.57	75.20
DWKD [24]	BraTS2019	72.96	88.66	78.97
MCTSeg [25]	BraTS2019	57.44	74.37	84.91
POD [26]	BraTS2019	80.10	83.60	84.30
TASCCNet [27]	BraTS2018	47.53	66.85	78.98
MKD [28]	BraTS2018	49.33	68.94	81.35
MTCM-KD [29]	BraTS2018	81.00	80.90	76.60
KD-Net [31]	BraTS2018	71.67	81.45	76.98
HAD-Net [33]	BraTS2019	39.80	66.70	87.90
PA-Seg (SCM) [34]	BraTS2019	–	–	85.60
3DKD-DiffuseNet	BraTS2017 (HG)	94.92	93.12	94.80
	BraTS2017 (LG)	93.22	91.45	93.80
	BraTS2018 (HG)	93.98	93.15	94.27
	BraTS2018 (LG)	93.02	91.22	93.22
	BraTS2019 (HG)	93.67	92.87	94.17
	BraTS2019 (LG)	92.22	90.32	92.57

Table 6

Generalizability comparison of our model against SOTA Diffusion-based models. The highest and second-highest results are presented in **bold** and underline, respectively.

Ref.	Model	Dataset	Dice (%)
[30]	FDiff-Fusion	BraTS2020	84.16
[38]	DDPM (ensemble)	BraTS2020	<u>90.9</u>
[39]	Diff-UNet	BraTS2020	85.35
[36]	HiDiff	BraTS2021	85.80
[37]	BerDiff	BraTS2021	89.7
[42]	MedSegDiff-L	BraTS2021	89.9
[43]	MedSegDiff-V2	BraTS2021	90.8
[56]	SegDiff	BraTS2021	89.3
[57]	D3PM	BraTS2021	84.88
Ours	3DKD-DiffuseNet	BraTS2020	92.01
		BraTS2021	<u>90.57</u>

performance comparable to or better than existing diffusion-based models. Table 6 reports the detailed comparison.

4.5. Evaluation on RAOS dataset

In Set A, which examines organs without surgery, our findings show that the proposed student model consistently outperforms the teacher model across various organs, with notable improvements in accuracy, precision, sensitivity, specificity, and Dice scores.

For well-defined organs like the bladder, kidneys, liver, and stomach, the student model achieved near-perfect accuracy, often surpassing 99%. For instance, the student model achieved 99.9% accuracy for the bladder and 99.5% for the left kidney, while the teacher model scored lower at 94.7% and 84.4%, respectively. This trend is consistent across Dice scores as well, with the student model consistently showing higher values, such as 97.01% for the right kidney, compared to the teacher's 94.01%.

For other organs, including the colon, femurs, and intestines, the student model again demonstrates better results, with accuracy ranging from 90% to 92% and Dice scores above 84%, compared to the teacher model's lower scores in the 80% range. Even in smaller or more challenging organs like the adrenal glands and the duodenum, the student model maintains superior performance, with the accuracy of the left adrenal gland reaching 76.8% compared to the teacher's 65.1%. Although the difference is less pronounced for organs such as the pancreas and rectum, the student model consistently holds an edge, with Dice scores around 81% for the pancreas and 85% for the rectum. Table 7 shows the detailed performance of Set A.

On Set B of the RAOS dataset, which evaluates the model's performance on samples after surgery without missing organs, the student model again consistently outperforms the teacher model across various metrics (see Table 8.). The performance for the bladder, for instance, is exceptional with an accuracy of 99.8% and a Dice score of 95.82%. This is followed closely by the results for the left kidney with a 99.7% accuracy and a Dice score of 96.46%. The accuracy for the right kidney is 99.5% with a Dice score of 92.12%. Similarly, for the liver, the model achieves a 99.9% accuracy and a Dice score of 96.78%. The model also performs well for the spleen, with a 99.6% accuracy and a Dice score of 91.11%. The accuracy for the intestine is 92.3% and a Dice score of 89.55%. However, the femurs show varied results. The accuracy for the left femur was only 76.1% with a Dice score of 72.02%; while for the right femur, the accuracy was 89.5% with a Dice score of 85.45%. Other organs, such as the adrenal glands and duodenum, have lower scores. The model achieves an 80.8% accuracy and a Dice score of 74.12% for the left adrenal gland, and a 76.4% accuracy with a Dice score of 72.54% for the right adrenal gland. The Dice score for the duodenum was only 72.0% for Dice, while the accuracy and Dice score for the colon are 81.1% and 83.23%, respectively. For the gallbladder, the model achieves a 79.8% accuracy and a Dice score of 80.12%. The model also demonstrates moderate performance for the rectum and pancreas, with accuracies in the range of 79.0% to 81.0% and Dice scores between 79.0% and 82.0%.

Set C of the RAOS dataset is used to evaluate the proposed student model's performance on organ segmentation when an organ is missing. The detailed result on Set C is present in Table 9. Similar to the previous sets (i.e., set A and set B), an accuracy of 99.9% is recorded for the bladder with a Dice score of 94.10%. The model also shows a strong performance for the spleen, achieving a 99.5% accuracy and a Dice score of 95.56%. For both kidneys, the model performs excellently, with a 99.5% accuracy and a Dice score of 91.20% for the left kidney and a 90.5% accuracy and a Dice score of 89.10% for the right kidney. The liver results follow closely, reaching a 96.5% accuracy and a Dice score of 94.89%.

The accuracy for the intestine is 90.8%, and the Dice score is 87.25%, while the accuracy and Dice score for the colon are 89.9% and 84.45%, respectively. The model maintains an accuracy of 86.8% with a Dice score of 82.82% for the esophagus, and the performance for the femurs shows consistency as well, with the left femur an 85.4% accuracy and a Dice score of 78.54%, and for the right femur an 87.4% accuracy and a Dice score of 81.38%.

The performance degrades for some organs, like the adrenal glands and gallbladder. The accuracy for the left adrenal gland reaches 75.4% with a Dice score of 71.12%, and the accuracy for the right adrenal gland is 76.2% with a Dice score of 64.23%. The accuracy for the gallbladder is only 59.1% with a Dice score of 59.85%. However, in comparing performance across all three sets, the proposed student model consistently demonstrates superior performance in accuracy, precision, and dice scores compared to the teacher model for most organs. Fig. 7 shows the segmentation output of the proposed student model (student diffusion).

4.6. Ablation tests

Inclusion and exclusion of preprocessing and loss functions. The preprocessing step (see Section 3.2) uses intensity-based ROI localization and slice selection to focus the model on relevant anatomical areas. The preprocessing reduces computational overhead by excluding irrelevant background and non-informative slices that do not contain tumors or organs. It also increases training and inference speed without affecting segmentation accuracy. This reduces the average epoch inference time from 135.4s to 92.8s. Additionally, we also note that knowledge distillation (KD) helps the lightweight student model learn better by mimicking the teacher's soft outputs. Without KD, the student's performance drops noticeably. The diffusion consistency loss

Table 7

Teacher (t/) and the proposed student (s/) model's performance on Set A: Without Surgery of the RAOS dataset. The best Dice scores are highlighted in **bold**, and the second-best scores are underlined. All results are reported in %.

Organs	SOTA Models			Our Study									
	SwinUNETR [58]	nnFormer [59]	nnUNet [54]	t/Dice	s/Dice	t/Acc	s/Acc	t/Prec	s/Prec	t/Sen	s/Sen	t/Sp	s/Sp
Adrenal (Left)	69.63 ± 18.09	55.99 ± 20.63	72.68 ± 18.84	70.19 ± 4.32	73.19 ± 4.54	76.8	65.1	66.8	74.3	62.8	68.8	68.9	81.2
Adrenal (Right)	69.1 ± 14.99	59.2 ± 16.48	71.82 ± 15.22	73.70 ± 4.85	75.70 ± 4.91	77.7	69.3	67.7	75.6	64.9	71.2	75.8	82.2
Bladder	93.73 ± 5.99	91.49 ± 9.56	94.86 ± 3.81	93.61 ± 1.14	95.61 ± 2.14	99.9	94.7	94.6	98.2	93.5	99.5	99.0	100
Colon	84.96 ± 8.12	81.65 ± 8.39	87.1 ± 8.39	81.10 ± 4.57	84.10 ± 4.93	92.1	86.1	84.6	89.7	81.7	85.9	88.5	93.5
Duodenum	68.72 ± 15.45	64.12 ± 14.67	72.05 ± 16.02	66.19 ± 4.79	69.19 ± 4.86	77.1	71.3	71.5	75.4	68.8	72.5	72.3	78.5
Esophagus	78.18 ± 8.84	71.6 ± 11.06	83.23 ± 5.51	83.47 ± 3.92	85.47 ± 4.01	88.2	83.4	82.4	87.2	81.5	84.5	84.2	90.8
Femur (Left)	91.79 ± 3.89	89.95 ± 3.95	84.99 ± 17.07	84.11 ± 4.45	87.11 ± 4.67	90.1	85.3	84.7	89.5	83.5	87.2	84.4	92.8
Femur (Right)	91.74 ± 3.98	89.18 ± 3.48	86.79 ± 9.11	82.34 ± 4.61	85.34 ± 4.74	91.8	84.6	83.5	89.5	81.1	86.3	92.1	94.3
Gallbladder	75.76 ± 18.25	71.07 ± 17.35	80.63 ± 17.36	79.12 ± 4.83	82.12 ± 4.92	85.6	77.3	76.4	84.2	76.4	81.2	89.5	89.0
Intestine	86.39 ± 6.28	84.05 ± 5.56	87.94 ± 6.54	86.32 ± 4.67	89.32 ± 4.79	92.9	83.1	81.1	91.8	81.1	89.5	89.5	93.5
Kidney (Left)	95.56 ± 1.11	94.57 ± 1.61	95.8 ± 1.18	91.53 ± 1.34	94.53 ± 3.34	99.5	84.4	84.1	98.7	83.5	99.2	87.8	100
Kidney (Right)	95.35 ± 1.32	94.53 ± 1.78	95.56 ± 1.36	94.01 ± 0.02	97.01 ± 2.02	99.1	84.1	83.4	98.6	82.6	97.5	99.1	100
Liver	96.4 ± 1.21	95.95 ± 1.23	96.65 ± 0.98	92.10 ± 0.83	95.10 ± 2.83	99.9	94.2	93.2	99.5	92.3	99.5	83.9	100
Pancreas	81.74 ± 8.45	77.16 ± 9.1	83.28 ± 8.2	78.10 ± 4.75	81.10 ± 4.83	88.3	85.0	84.5	87.3	84.2	85.5	93.2	88.7
Prostate	68.73 ± 37.77	84.75 ± 17.81	93.01 ± 6.94	87.45 ± 3.93	90.45 ± 4.21	96.4	90.8	89.8	95.8	89.5	94.8	86.8	96.6
Rectum	80.96 ± 10.66	71.93 ± 12.12	83.1 ± 11.16	81.92 ± 4.59	84.92 ± 4.68	88.1	84.6	83.9	87.2	82.5	84.6	93.3	90.5
Seminal Vesicle	63.63 ± 37.04	77.48 ± 23.38	88.23 ± 14.58	87.11 ± 4.75	90.11 ± 4.89	93.2	89.5	88.9	92.6	87.3	91.1	97.3	94.8
Spleen	95.72 ± 0.97	94.69 ± 1.51	95.93 ± 1.09	90.12 ± 1.67	93.12 ± 3.67	97.4	91.0	90.5	96.6	91.5	95.1	97.4	98.2
Stomach	92.04 ± 3.56	90.78 ± 3.59	92.77 ± 3.69	90.97 ± 3.12	92.97 ± 4.25	98.1	92.6	91.9	97.6	91.3	96.5	96.9	99.3

Table 8

Teacher (t/) and the proposed student (s/) model's performance on Set B: After Surgery (w/o missing organ) of the RAOS dataset. The best Dice scores are highlighted in **bold**, and the second-best scores are underlined. All results are reported in %.

Organs	SOTA Models			Our Study									
	SwinUNETR [58]	nnFormer [59]	nnUNet [54]	t/Dice	s/Dice	t/Acc	s/Acc	t/Prec	s/Prec	t/Sen	s/Sen	t/Sp	s/Sp
Adrenal (Left)	71.7 ± 9.77	57.28 ± 17.31	75.17 ± 9.2	70.12 ± 4.65	74.12 ± 4.85	74.8	80.8	74.3	78.2	71.7	75.5	79.7	82.3
Adrenal (Right)	68.53 ± 13.18	60.56 ± 15.98	71.0 ± 14.42	68.54 ± 4.72	72.54 ± 4.58	71.1	76.4	69.8	74.3	67.9	71.2	75.4	79.2
Bladder	92.68 ± 10.85	90.87 ± 12.83	93.12 ± 10.75	92.82 ± 2.75	95.82 ± 3.75	93.9	99.8	91.8	99.3	94.4	99.6	98.3	100
Colon	80.98 ± 9.98	76.98 ± 10.06	84.06 ± 10.1	79.23 ± 4.81	83.23 ± 4.66	81.1	89.5	82.5	87.0	81.2	85.5	83.5	88.7
Duodenum	69.08 ± 12.88	65.33 ± 10.83	70.63 ± 11.85	68.00 ± 4.89	72.00 ± 4.55	72.4	76.9	72.3	75.4	69.4	72.5	72.3	77.9
Esophagus	77.65 ± 7.86	71.46 ± 9.8	81.59 ± 4.94	78.00 ± 4.77	82.00 ± 4.91	82.4	87.8	81.5	86.2	79.4	84.1	83.2	88.7
Femur (Left)	91.44 ± 4.35	89.49 ± 4.3	70.52 ± 32.47	69.02 ± 4.66	72.02 ± 4.92	68.6	76.1	70.3	74.2	68.3	71.8	74.8	78.6
Femur (Right)	91.85 ± 4.15	88.91 ± 3.36	83.35 ± 10.91	81.45 ± 4.59	85.45 ± 4.62	80.5	89.5	83.6	87.2	79.9	85.9	83.1	88.4
Gallbladder	74.6 ± 14.6	60.22 ± 25.56	78.03 ± 14.33	76.12 ± 4.71	80.12 ± 4.88	79.8	83.1	78.4	81.5	81.5	79.2	79.5	83.9
Intestine	84.37 ± 7.48	81.86 ± 6.68	86.55 ± 6.45	85.55 ± 4.93	89.55 ± 4.77	87.7	92.3	86.7	91.2	75.7	89.5	88.5	93.1
Kidney (Left)	95.33 ± 1.22	93.93 ± 1.66	94.46 ± 5.04	93.46 ± 1.04	96.46 ± 3.04	96.4	99.7	93.4	99.1	83.5	99.4	97.6	100
Kidney (Right)	95.39 ± 0.98	94.48 ± 1.6	94.78 ± 4.25	89.12 ± 4.86	92.12 ± 4.79	89.2	99.5	89.6	98.1	93.4	97.3	98.4	100
Liver	96.36 ± 1.08	95.75 ± 1.65	96.06 ± 2.57	93.78 ± 2.57	96.78 ± 3.57	88.7	99.9	87.3	99.6	92.2	99.5	91.2	99.1
Pancreas	81.87 ± 8.46	77.08 ± 9.04	84.11 ± 6.65	79.53 ± 4.89	82.53 ± 4.65	81.2	89.5	84.2	87.9	90.8	85.5	85.2	89.1
Prostate	75.28 ± 26.42	76.48 ± 22.09	87.31 ± 10.16	87.31 ± 4.41	90.31 ± 4.53	85.9	94.2	88.5	93.7	81.5	91.5	90.8	95.1
Rectum	77.53 ± 10.62	73.95 ± 10.35	80.25 ± 9.49	79.19 ± 4.73	82.19 ± 4.92	79.2	85.5	79.8	84.2	86.9	82.1	81.4	86.1
Seminal Vesicle	66.56 ± 33.23	57.79 ± 35.3	79.99 ± 23.38	77.01 ± 4.59	80.01 ± 4.75	79.0	82.8	78.5	81.5	77.9	80.3	81.5	85.2
Spleen	95.28 ± 1.85	94.64 ± 1.44	95.32 ± 2.41	88.11 ± 2.41	91.11 ± 4.47	89.4	99.6	94.3	99.2	76.6	99.3	97.5	100
Stomach	90.1 ± 8.43	89.35 ± 6.77	91.94 ± 4.46	90.12 ± 4.77	93.12 ± 4.92	88.7	98.2	90.8	97.5	93.5	96.4	93.5	98.5

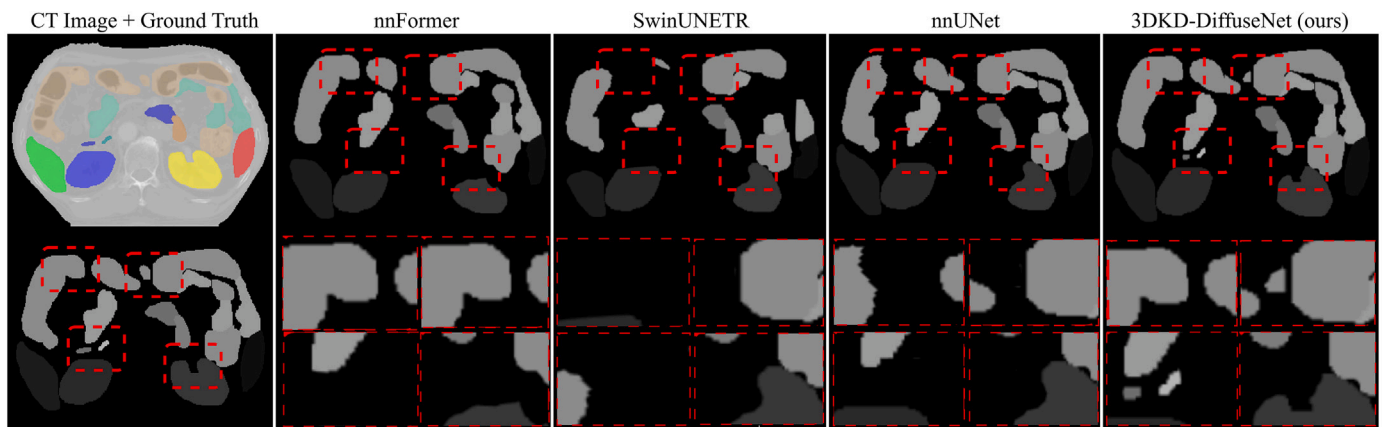


Fig. 7. Segmentation output on the RAOS dataset. The first column illustrates a sample image with a ground truth mask provided in the second row. The rest of the columns compare SOTA models' predicted masks. The highlighted regions in the first row (second to fifth column) are zoomed in on the following row.

Table 9

Teacher (t/) and the proposed student (s/) model's performance on Set C: After Surgery of the RAOS dataset. The best Dice scores are highlighted in **bold**, and the second-best scores are underlined. All results are reported in %.

Organs	SOTA Models			Our Study									
	SwinUNETR [58]	nnFormer [59]	nnUNet [54]	t/Dice	s/Dice	t/Acc	s/Acc	t/Prec	s/Prec	t/Sen	s/Sen	t/Sp	s/Sp
Adrenal (Left)	66.8 ± 15.86	51.16 ± 26.37	72.46 ± 15.78	66.12 ± 18.65	71.12 ± 21.54	75.4	79.4	75.5	78.2	70.9	74.9	78.3	80.3
Adrenal (Right)	59.85 ± 26.64	53.73 ± 26.0	65.65 ± 25.07	61.23 ± 24.45	64.23 ± 27.25	76.2	73.1	71.5	72.4	66.7	69.7	72.1	74.0
Bladder	91.42 ± 10.14	89.01 ± 12.38	92.57 ± 9.36	90.15 ± 3.55	94.10 ± 5.09	93.9	99.9	96.5	99.5	94.8	99.8	97.8	100
Colon	80.25 ± 9.67	80.08 ± 8.64	84.9 ± 10.69	81.52 ± 5.52	84.45 ± 8.14	89.9	92.0	84.2	91.1	83.9	88.9	89.5	93.5
Duodenum	58.16 ± 18.68	57.8 ± 16.81	64.79 ± 17.55	62.10 ± 11.52	63.55 ± 16.30	85.4	87.2	73.3	86.2	81.5	84.5	87.3	87.7
Esophagus	73.48 ± 17.13	68.98 ± 15.8	80.51 ± 9.84	78.41 ± 7.14	82.82 ± 10.22	86.8	87.8	84.3	86.5	79.4	83.2	86.3	88.3
Femur (Left)	90.26 ± 11.23	87.94 ± 11.66	79.64 ± 22.58	77.52 ± 22.51	78.54 ± 21.30	85.4	87.7	72.4	86.5	79.6	82.9	84.8	88.1
Femur (Right)	89.4 ± 13.38	86.77 ± 12.71	80.67 ± 21.41	74.12 ± 20.42	81.38 ± 18.58	87.4	88.0	84.6	87.6	84.3	87.3	85.2	88.2
Gallbladder	40.19 ± 40.99	42.32 ± 40.04	54.69 ± 42.8	55.12 ± 31.68	59.85 ± 34.25	59.1	62.2	78.8	69.9	65.7	69.7	63.7	65.7
Intestine	82.81 ± 8.27	81.63 ± 8.34	86.07 ± 7.65	84.15 ± 6.33	87.25 ± 8.92	90.8	93.5	89.5	92.5	88.2	91.2	90.8	93.8
Kidney (Left)	90.24 ± 19.04	90.91 ± 15.83	90.95 ± 18.92	88.15 ± 4.32	91.20 ± 7.45	96.5	99.5	96.5	99.7	96.5	99.5	99.7	100
Kidney (Right)	79.28 ± 34.01	89.16 ± 20.25	89.36 ± 21.98	85.45 ± 8.12	89.10 ± 10.45	90.5	92.7	95.2	91.8	87.0	90.3	91.4	93.4
Liver	94.82 ± 9.75	94.69 ± 8.72	95.35 ± 8.73	92.67 ± 1.56	94.89 ± 3.21	96.5	99.2	97.5	98.6	95.1	99.1	98.5	100
Pancreas	77.32 ± 13.66	72.47 ± 14.8	81.05 ± 12.36	81.15 ± 8.35	83.65 ± 10.52	85.5	88.2	85.5	87.9	82.2	85.2	86.8	88.8
Prostate	53.65 ± 41.8	75.99 ± 30.08	80.23 ± 31.1	78.12 ± 15.15	81.36 ± 18.25	82.0	85.3	91.1	84.2	79.1	82.1	82.8	85.3
Rectum	73.02 ± 22.85	65.32 ± 22.85	77.26 ± 21.4	77.45 ± 13.10	79.90 ± 15.10	81.5	84.5	82.0	83.7	78.2	81.2	84.7	85.7
Seminal Vesicle	39.25 ± 39.99	69.11 ± 35.28	68.13 ± 38.82	66.12 ± 27.11	69.45 ± 30.11	77.4	81.7	79.5	80.4	74.6	78.6	79.3	82.3
Spleen	94.78 ± 2.34	93.67 ± 3.16	95.37 ± 2.08	93.12 ± 3.12	95.56 ± 1.78	96.3	99.5	96.6	99.1	96.4	99.4	98.6	100
Stomach	88.63 ± 9.31	88.25 ± 8.86	90.75 ± 7.93	90.12 ± 4.95	92.98 ± 6.95	96.4	98.2	94.7	97.5	93.5	96.5	95.5	98.0

Table 10

Ablation study on the BraTS2017 dataset showing the effect of including/excluding preprocessing steps and different loss function combinations in training the student model. Reported metrics include Dice score and computational efficiency measures. Here, the inference time is presented in average seconds per epoch, noted in the Inference column. The total epochs for this experiment were 100.

Model Variant	ROI Pre.	KD (L_K)	Diff. (L_{diff})	CE (L_C)	Inference↓	Dice
Only Student	✗	✓	✗	✓	124.8s	79.05
- w/o ROI Preprocessing	✗	✓	✓	✓	135.4s	83.76
- w/o KD	✓	✗	✗	✓	103.1s	84.91
- w/o Diffusion Consistency	✓	✓	✗	✓	87.9s	87.23
Full 3DKD-DiffuseNet	✓	✓	✓	✓	92.8s	90.15

improves the model's robustness by encouraging it to handle noisy or uncertain inputs effectively. The detailed results are reported in Table 10.

Weights tuning for loss functions. We evaluated various weighting schemes for the loss components during training of the student model. The total loss combines cross-entropy (L_C), knowledge distillation via KL divergence (L_K), and diffusion consistency (L_{diff}) terms weighted by α , β , and γ , respectively (see Eq. (14)). The weights were tuned using a grid search under the constraint that α , β , and γ are constrained such that their total is 1 (except for the baseline). The experiments showed that maintaining a relatively high α was essential, since segmentation loss (L_C) provides the primary signal for accurate boundary prediction. The distillation weight β was most effective in the range 0.2–0.3, which enabled the student to benefit from teacher guidance without overfitting. On the other hand, the diffusion consistency weight γ was used as a regularizer, with small values (0.1–0.2) improving robustness, while larger values reduced training stability. It is worth noting that, unlike a standard segmentation task where only the cross-entropy loss would typically be prioritized, in our case, all three components were necessary to achieve optimal performance, as reflected in the reported results on BraTS2017 (see Table 11). When we trained the student using only cross-entropy, the Dice score was 0.790. Adding knowledge distillation improved it to 0.849. Including the diffusion consistency loss further increased performance, and the best results came with weights [0.5, 0.3, 0.2], reaching a Dice of 0.901.

Model's parameters count. Our method, even with the diffusion module included, maintains a relatively low parameter count, which is substantially fewer than models like MedSegDiff-V2 [43] or SwinUNETR [60]. The student model contains only 21M parameters and is significantly smaller than the teacher (64M) and many SOTA architectures, including SwinUNETR (138M), UNETR (104M), and TransUNet (96M). This is primarily because we decouple the heavy generative

Table 11

Effect of different weighting settings for the three loss terms. The total loss (equivalent to 1) combines CE loss, KL divergence, and diffusion consistency terms weighted by α , β , and γ , respectively. Dice scores (%) demonstrate how tuning these hyperparameters impacts segmentation performance.

Method	α	β	γ	Dice
Teacher (base)	0	0	0	0.783
Student (L_C only)	1	0	0	0.790
+ KD	0.7	0.3	0	0.849
+ KD + Diffusion	0.4	0.4	0.2	0.881
+ KD + Diffusion	0.5	0.3	0.2	0.901
+ KD + Diffusion	0.6	0.3	0.1	0.885

nature of traditional diffusion models and instead integrate diffusion as an auxiliary denoising process, rather than modeling the entire data distribution. Additionally, unlike our teacher model (64M), our student model also avoids complex attention mechanisms and deep transformer layers that significantly increase model size in other architectures. Through KD, the student retains high segmentation performance despite the reduction in depth and parameters. In particular, the student network uses a shallow 2-depth architecture guided by diffusion consistency with KD. Although the diffusion module adds a small number of extra parameters (+3M), the resulting performance gain justifies this tradeoff. Table 12 compares the parameter counts of our proposed model against existing segmentation architectures reported in the literature.

5. Discussion

The primary objective of our study was to reduce computational time while enhancing the accuracy of medical imaging models. Our key findings indicate that converting CT or MRI images into an array

Table 12

Comparison of model sizes (parameter counts) between our approach and representative segmentation architectures. Our final model (last row) includes the diffusion process in the student network.

Ref.	Model	Param(M)
Hatamizadeh et al. [60]	SwinUNETR	138
Hatamizadeh et al. [61]	UNETR	104
Chen et al. [62]	TransUNet	96
Wu et al. [43]	MedSegDiff-V2	46
Wu et al. [42]	MedSegDiff	25
Amit et al. [56]	SegDiff	23
Isensee et al. [54]	nnU-Net	16
3DKD-DiffuseNet	Teacher	64
	Student	19
	+ Diffusion	21

format through organ-specific intensity thresholding significantly reduced computational time without compromising critical 3D modality features. This preprocessing step improved the model's efficiency while maintaining essential elements of the medical images.

Unlike prior work that only uses KL divergence for knowledge distillation, our approach leverages a discriminative diffusion framework to improve generalization. This involves a forward diffusion step where Gaussian noise is added to the input volume, followed by a reverse step where the student's encoder acts as a denoiser to reconstruct the original input. This process is supervised by a novel diffusion consistency loss that directly regularizes the student's feature space and enhances its ability to learn robust, noise-invariant representations.

Furthermore, incorporating attention mechanisms into the teacher model decoder, along with our diffusion-augmented student model, led to a significant increase in model accuracy. The architectural synergy between the teacher's attention-guided feature refinement and the student's diffusion-driven robustness demonstrates the efficacy of this hybrid design, as evidenced by the consistent performance improvements across both datasets.

The interpretation of our results suggests that the proposed framework is efficient and robust, delivering superior performance on two distinct categories of datasets. The computational gains of the framework are primarily achieved through two components: ROI-based preprocessing and knowledge distillation. The preprocessing pipeline reduces unnecessary overhead by focusing on relevant ROIs, applying intensity thresholding to exclude irrelevant background, and removing slices that do not contain tumors or organs. Furthermore, knowledge distillation allows the student model to inherit the performance of a larger teacher network (4 depths) while using a shallower architecture with only 2 depths. Our findings show that including preprocessing reduces the average epoch inference time from 135.4s to 92.8s. Also, with diffusion, the proposed student model requires fewer parameters (i.e., 21M) compared to several SOTA U-Net- and diffusion-based models, such as SwinUNETR (138M), TransUNet (96M), SegDiff (23M), and MedSegDiff (25M). Furthermore, the computational efficiency exhibited by the framework positions it as an efficient solution for real-time medical image processing applications.

Compared to prior research in multi-organ imaging, our results significantly surpass the performance of existing methods, emphasizing the superiority of our approach in both accuracy and computational efficiency. SOTA studies [16–23] using knowledge distillation have reported Dice scores of 73.60% for brain tumor segmentation [22], 78.57% for the same task in another study [21], and 85.91% for multi-organ segmentation in the WORD dataset [23]. In contrast, our study shows a significant improvement. For the 2017–2019 Brain Tumor Segmentation data sets, the student model achieved maximum Dice scores for high-grade tumors in FLAIR images, outperforming the teacher model by 3 to 5% in all modalities. It also achieved strong accuracy for low-grade tumors, with Dice scores of up to 97.8% for whole tumor

classification. Similarly, for the RAOS dataset, the student model consistently outperformed the teacher model, achieving accuracies exceeding 99% for organs such as the bladder (99.9%) and kidneys (99.5%). Critical organs like the adrenal glands saw Dice score improvements of 3%–6%, maintaining robust results even in challenging cases. Furthermore, inference time analysis revealed a 2–3 times reduction in computational time due to optimized preprocessing, enabling real-time feasibility.

However, the study does have a few limitations: firstly, we only relied on public benchmark datasets for model training and evaluation. Real-time hospital data could validate the practical applicability of our framework, as public datasets do not always have the complexity and variability of real-time medical images. Second, although the proposed student model achieved excellent overall performance on the RAOS dataset for major organs, certain smaller and anatomically variable organs, including the adrenal glands, gallbladder, and femurs showed relatively lower scores (for instance, left adrenal: 71%–74%, right adrenal: 64%–76%, gallbladder: 59%–82%, left femur: 72%–78%, right femur: 81%–85% across the 3 sets). These differences reflect the inherent challenges of segmenting small and thin structures, organs with high shape variability, which tend to present consistent difficulties across segmentation models (including SwinUNETR, nnFormer, and nnUNet) as well. Additionally, fewer voxels for smaller organs reduce the influence of the loss function during training. Post-surgical anatomical changes in Sets B and C further increase segmentation complexity.

Despite the limitation, the proposed framework has the potential to substantially improve the speed and accuracy of image analysis in clinical settings, thereby facilitating faster diagnoses and more reliable medical predictions. The reduction in computational time could also enable a more efficient deployment of machine learning models in resource-constrained environments. The lower performance observed in the RAOS subsets in multi-organ segmentation can be mitigated by incorporating uncertainty-guided refinement to prioritize regions with higher prediction ambiguity. In addition, integrating 3D organ-specific shape priors can improve the model's ability to accurately delineate challenging boundaries. For the brain tumor segmentation task, in our current study, we have primarily focused on subregion segmentation in adult patients, as defined by the non-pediatric benchmark. In future research, we will prioritize integrating real-time clinical datasets to assess the framework's performance under real-world conditions. Expanding the dataset to include diverse imaging modalities and patient demographics will further validate the model's robustness and generalizability. Particularly, we aim to extend the framework to pediatric datasets, which present distinct challenges due to differences in tumor types, morphology, and imaging characteristics, and we also plan to include label-wise evaluation to provide a more detailed assessment of segmentation performance across both adult and pediatric populations.

6. Conclusion

In conclusion, this study presents an innovative methodology to enhance the computational efficiency and accuracy of medical image analysis. By implementing organ-specific intensity thresholding, we significantly reduced computational time while preserving essential three-dimensional features in volumes of the BraTS MRI and RAOS CT datasets. The standardization of the voxel intensities and the normalization of the CT data contributed to the consistency of the various imaging modalities. In addition, we developed a resource-efficient method for identifying the region of interest, along with a novel three-dimensional attention-based teacher model complemented by a diffusion-enhanced student model, which significantly improved the overall performance of the model. The robustness of our framework was demonstrated across multiple datasets, providing a scalable solution for real-time medical imaging.

CRediT authorship contribution statement

Mohaimenul Azam Khan Raiaan: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Md. Abdur Rahman:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Sami Azam:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Methodology, Formal analysis, Conceptualization. **Kheng Cher Yeo:** Writing – review & editing, Writing – original draft, Visualization, Validation. **Yakub Sebastian:** Writing – review & editing, Writing – original draft, Visualization, Validation. **Mirjam Jonkman:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision.

Ethics statement

This research did not involve the use of human participants, animals, or clinical trials. All datasets utilized in this study are publicly available and fully anonymized, ensuring compliance with ethical standards. No additional ethical approval was required for this research.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

This study utilized two distinct 3D medical segmentation tasks datasets: (1) brain tumor segmentation using BraTS benchmark datasets (BraTS2017–2019), with additional validation on BraTS2020 and BraTS2021 to assess generalizability; and (2) abdominal organ segmentation using the RAOS dataset. The BraTS benchmark datasets are publicly available, and the RAOS dataset is available upon request from the respective dataset owners. (<https://github.com/Luoxd1996/RAOS>).

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